



R V College of Engineering
Department of Computer Science and Engineering
Improvement CIE: Question Paper

**Subject :
(Code)**

COMPILER DESIGN (CS363AI)

Semester : 6th BE

Date : JUNE 2025

Duration : 120 minutes

Staff : DD/SB/SNM/SMS/PK

Name :

USN :

**Section :
A,B,C,D, ISE**

Si. No	Part A Quiz Questions	M	B T	Co
1.	What is a short circuit code? Give an example.	1	1	CO3
2.	Write the control flow translation of the following statement: If($x < 100 \parallel x > 200 \ \&\& \ x \neq y$) $x = 2$;	2	4	CO4
3.	Construct a DAG for the following basic block: $T1 = A + B$ $T2 = C + D$ $T3 = E - T2$ $T4 = T1 - T3$	2	4	CO4
4.	_____ function inserts I as the target label for each of the instructions on the list pointed to by p in one pass code generation using backpatching.	1	2	CO3
5.	Write the backpatching SDT for the following production: $B \rightarrow B1 \ \&\& \ M \ B2$	2	2	CO4
6.	What is the cost of the instruction LD R1, *100(R2)? Justify your answer.	2	3	CO4
Si. No	Part B Test Questions	M	B T	Co
2.	Write the different techniques for optimizing basic blocks with examples.	10	4	2
3.	Consider the following source code: for i from 1 to 10 do for j from 1 to 10 do C[i, j] = A[i, j] + B[i, j]; Write the TAC considering the array sizes as 10X 12 where each element takes 8 bytes of memory space. Identify the leaders and form the basic blocks Draw the flow graphs and identify the loops if one exists	10	4	2
4.	Explain the main issues in code generation with examples for each of the issues in detail.	10		3
5.	How the following basic block can be optimized? 1. a = 10	10	2	3

	2. b=4*a 3. t1=i*j 4. c=t1+b 5. t2=15*a 6. d=t2*c 7. e=i 8. t3=e*j 9. t4=i*a 10. c=t3+t4 Explain with examples Peephole optimization.			
5.	Using backpatching manipulate the Truelist and the Falselist at the root node of the annotated parse tree, which should be constructed for the expression: a<b && b>c && a!=d.	10	2	4